

Enhanced Signal Processing for Single-Reader/Dual-Track Reading (SRDR) Bit-Patterned Magnetic Recording Systems

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To eliminate the need for a narrow-track reader in a high-track-density BPMR system, this work investigates encoding schemes and dual-track simultaneous reading, pointing to the use of a wide-track reader. High two-dimensional (2-D) interference constitutes a significant obstacle. To avoid 2-D interference, two enhanced signal processing methods are introduced: 1) a simple modulation code at a rate of 3/4 and 2) a modified Viterbi detection scheme. Since data from two tracks is read simultaneously, the transfer speed is twice that of single-track reading. Moreover, the simulation results show that our proposed system performs better than single-track reading in terms of bit-error rate at the same user densities (UDs), especially at ultra-high UD.

Index Terms—Signal processing, single reader/dual track reading, bit-patterned magnetic recording, 3/4 modulation code, modified Viterbi algorithm.

I. INTRODUCTION

CURRENTLY, the hard disk drive (HDD) industry is primarily focused on increasing storage capacity and improving recording performance. To achieve this, new ultrahigh-density magnetic recording technologies, such as bit-patterned magnetic recording (BPMR) [1,2], have been proposed. BPMR seeks to surpass the limits of conventional perpendicular magnetic recording [3], which is restricted to around 1.0 Terabit per square inch (Tb/in²). One of the advantages of BPMR is its reduced sensitivity to transition noise, as each data bit is stored in an isolated single-domain magnetic island, which is surrounded by a nonmagnetic region. Therefore, BPMR technology [2] is expected to reach an areal density (AD) of 4.0 Tb/in², or more. However, to obtain higher ADs, the bit-islands must be very close together (narrow track width) and, since the reader is normally wider than the track pitch, intertrack interference (ITI) and intersymbol interference (ISI) effects are also unavoidably increased [4]. These 2-D interference effects severely degrade the system performance, especially at high ADs.

In this paper, we present a single-reader/dual-track reading (SRDR) scheme that enables simultaneous reading of two neighboring tracks with a wide-track reader, as illustrated in Fig. 1, thereby mitigating the need for a narrow-track reader. Strong 2-D interference or cross-talk noise between neighboring bits/tracks is the major obstacle. We then present a rate-3/4 modulation code, which is used to encode the user bits of two neighboring tracks. Moreover, we propose a modified Viterbi detector in which the Trellis branches are designed according to the reading scheme. A computer simulation shows that our proposed system can achieve better bit-error rate (BER) performance than single-track reading at ultra-high ADs.

II. CHANNEL MODEL

A block diagram of the system with SRDR BPMR is shown in Fig. 2. An input sequence $\{u_k\}$ is encoded by a proposed rate-3/4 modulation code that produces two recorded sequences

$\{a_k^{\text{up}}, a_k^{\text{lw}}\}$, where up and lw denote the upper- and lower-track, respectively, which are then recorded onto a medium. Note that all recorded bits are assumed to be perfectly written, which means their magnetization is the same as the recorded bits. The patterned media considered in this study consist of a regular array of circular islands with a diameter (D) of 10 nm. The AD of the patterned media is determined by the bit period T_x and track pitch T_z . The soft-magnetic underlayer (SUL) had a thickness of 300 nm, the nonmagnetic interlayer (IL) was 15 nm thick, the recording layer (RL) was 10 nm thick, and the head-medium spacing (HMS) was 5 nm [5].

In the reading process, two data tracks are read by a single reader simultaneously, where the single reader is placed between the upper- and lower-tracks, as shown in Fig. 1. The readback signal r_k of the upper- and lower-tracks can be easily generated by sampling the readback signal at a rate $t = kT_x$. Thus, we obtain readback sequences. The 1-D equalizer considered in this study optimizes the readback sequence using an MMSE criterion to meet the fixed 1-D generalized partial response (GPR) target. After equalization, the modified 1-D Viterbi detector detects the equalized sequences. Modified 1-D Viterbi detectors that depend on their modulation code are employed to determine the most likely recorded sequences. The obtained sequence is then decoded to generate the estimated binary input sequence, $\hat{u}_k$. Finally, the BER is used to evaluate the proposed system's performance.

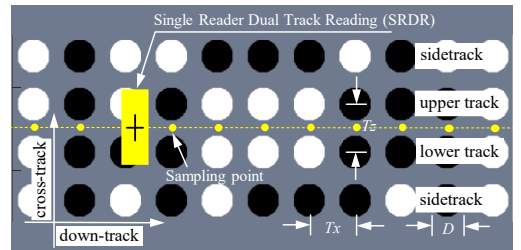


Fig. 1. Regular bit-patterned medium with a single reader used to read dual data tracks simultaneously.

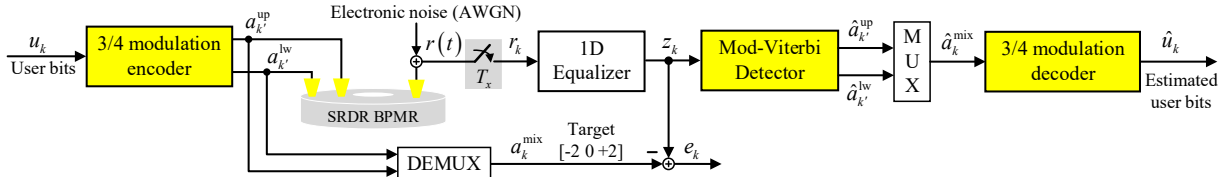


Fig. 2. Block diagram of the proposed SRDR BPMR system, which consists of a rate-3/4 modulation encoder/decoder and modified 1-D Viterbi algorithms.

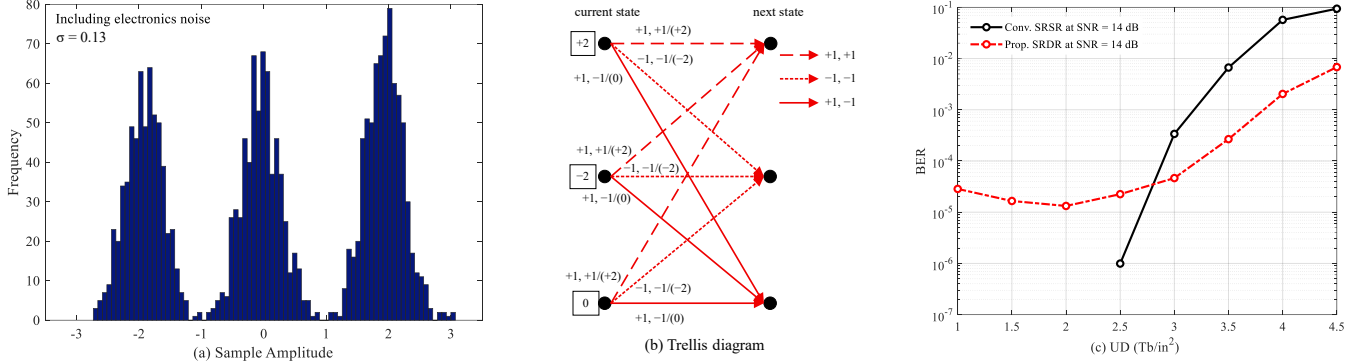


Fig. 3. PRML channel enables two-track reading with a wide reader, as illustrated by (a) equalizer output distribution, (b) a Trellis diagram where the path notation X/Y denotes the decoded data X alongside the readback amplitude Y , and (c) BER performance of the proposed system versus a conventional system at several UD.

III. PROPOSED TECHNIQUES

To design suitable codewords for the SRDR BPMR system, we first investigated the normalized error percentage and found that the recorded data patterns should not include both ‘-1, +1’ or ‘+1, -1’ in the upper- and lower-tracks. These data patterns are difficult to distinguish in the SRDR BPMR system. Here, our coding design partitions the input sequence into 4-bit packets, yielding 16 possible patterns. Hence, we choose the optimal 4-bit data pattern, including ‘+1, +1’, ‘-1, -1’, and ‘+1, -1’, excluding ‘-1, +1’, to record on the upper- and lower-tracks, respectively, to prevent the aforementioned issue. Moreover, using this rate-3/4 modulation code, when we sample the readback signal at the rate $t = kT_x$, as shown in Fig. 1, we obtain sample amplitudes of around +2, 0, and -2 for ‘upper- and lower-track’, as shown in Fig. 3(a). Fig. 3(a) shows three distinct data groups, making it easier to identify which group corresponds to ‘+1, +1’, ‘+1, -1’, or ‘-1, -1’ using a simple threshold detector. However, as UD increases, the detector’s effectiveness may diminish. Therefore, we also propose a modified Viterbi detector that effectively identifies the bits produced by the encoding scheme.

To enhance the performance of SRDR detection in the BPMR system, we propose dual-track PRML detection, in which the conventional Trellis diagram is modified to reflect the new state diagram, as shown in Fig. 3(b). The conventional Trellis diagram has four stages in this PR-2 situation [6], yielding eight paths in total. In contrast, the branch metric of our modified Trellis diagram is computed from all possible transitions of the recorded magnetization at each sampling point. Therefore, the complexity of our proposed detector will be reduced to three stages with nine paths in total, and it can detect two bits simultaneously, implying a transfer speed twice that of single-track reading.

IV. RESULTS AND CONCLUSIONS

In this study, we assess the BER performance of different recording schemes under several user densities. These consist of 1) conventional single-reader/single-track reading (Conv. SRSR) and 2) the proposed single reader/dual-track reading (Prop. SRDR). The simulation results show that our proposed detection technique improves BER performance compared to the Conv. SRSR method when the UD exceeds 2.8 Tbit/in², as shown in Fig. 3(c). In particular, the proposed system remains tolerant of severe media noise and 2-D interference even at UD of 4 Tbit/in² and above. This outcome demonstrates that the proposed rate-3/4 modulation code and modified Viterbi detector can effectively implement single-reader/dual-track reading in BPMR systems.

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